

PROFESSIONAL – Plant Engineering

To obtain position in organization by utilizing knowledge and skill in the best possible ways and learn continuously the best practice to attain challenging position in an organization. To lead the team for enhancing the growth of the organization.

Core Competencies

Instrumentation •Industrial Automation •Operations/Maintenance •Procurement •Technical Evaluation •Breakdown Analysis •Cost Optimization •PLC/SCADA •DCS •DAQ System •NI system •Trouble shooting •Process Enhancement •Preventive maintenance •Predictive maintenance •Safety •Team Management •Coordination •Innovation •Improvement •Project Engineering •Planned maintenance (spare parts and man power management) •Parts Localization.

KEY SKILLS

- Expertise in installation and calibration of Instrumentation like as Flow transducer (mass flow, coriolis, turbine type), pressure transducer, temperature sensor (RTD, thermocouple, thermistor), speed transducer (encoder, magnetic pulse, hall effect type), torque transducer (flange type, load cell based), LVDT, Level transducer, accelerometer, gas analysers (CO, CO₂, NOX, THC, O₂), control valves.
- Expertise in selection of transducer as per the application and signal wiring techniques.
- Expertise in conversion of analog signal as per availability of analog voltage or current channels.
- Worked on universal protocols such as OPC, Wireless communication, Profibus, Industrial Ethernet, Modbus, Ethernet/IP, Foundation Fieldbus.
- Rich Experience in Project Management with Initiation, Planning, Execution, Monitoring & Control and Closing.
- Well versed in developing logic programs and SCADA based system to monitor & control engine testing equipments like CCU, FCU, transient Dyno, CAHU, pumps of building for filling the water tank with auto on/off on time basis & start/stop.
- Expertise to control the system in closed loop through PID.
- Configuring and fault diagnosis of DCS, PLC, SCADA and control loops.
- Design SCADA and PLC logic development, HMI display, MATLAB, Arduino, C++, Proteus.
- Experienced on pneumatic and Hydraulic systems

PROFESSIONAL EXPERIENCE**INTERNATIONAL CENTRE FOR AUTOMOTIVE AND TECHNOLOGY****March 2015-till date****Manesar, Haryana** [ICAT - International Centre for Automotive Technology, Manesar India](#)

International Centre for Automotive Technology (ICAT) is an Automotive Testing and R & D centre which is one of the flourishing business hubs in North India. It is the apex body of NATRIP Implementation Society (NATIS), Govt. of India in northern region to provide homologation facility to all Vehicles and automotive parts.

Assistant Manager

- Supervision, co-ordination and managing the day-to-day and round the clock operation and maintenance of all specialized services and utilities.
- Planned & supervised work progress on a daily, weekly and monthly basis.
- Study of P&ID, Electrical, Hydraulic and pneumatic circuit drawings to find out the root cause of problem in machine and discussions about control strategy based out of them.
- Expertise in managing maintenance operations for reducing breakdown / downtime & enhancing operational effectiveness of equipment.

- Project Planning and Execution, estimation, tracking & implementation of project plans within preset budgets and deadlines and coordinating projects for the set-up facilities.
- Preparing maintenance/services schedules of various process control equipments.
- Expertise in ensuring highest standards of quality, safety, reliability, integrity & cost-efficiency through engineering excellence and carrying out instrumentation activities & enhancing operational effectiveness of equipment
- Instrumental in managing end-to-end management of multiple projects, monitoring progress of project as per schedule and ensuring timely completion & delivery of project to the client.
- Review and modification of Hook up drawing, P & ID, Electrical Wiring drawing and panel drawings
- Effective utilization of manpower for breakdown and its analysis.
- Conducting technical trainings, proof of concept testing, Team Co-ordination.
- Performed root-cause analysis of breakdown for taking corrective actions to prevent any reoccurrence of issues.
- Done quality analysis to improve the efficiency of the machine.
- Localization of vendor to reduce monopoly of OEM so that it helps in cost reduction product

ACHIEVEMENTS

- Done automation for salt spray chamber using PLC
- Designed the controller for coolant conditioning unit with PC interfacing.
- Done automation for fill fuel in 4 tanks simultaneously with proper interlocking.
- Design setup to check the functioning of Sierra CP control panel electronics cards and control valve.
- Design setup for thermal shock cycle.
- Made alteration in CAHU unit to achieve 38 deg Celsius
- Designed Heated Line compatible for all emission analysers (AVL, HORIBA)
- Designed the heater for HORIBA analyser. (not available in Indian Market)

KLB INSTRUMENT PVT. LTD, DELHI

Jun'12- Feb'15

KLB Instruments is the dealer exclusive for high temperature and pressure batch Reactors of PARR Instruments, U.S.A
[\(http://www.parrinst.com/\)](http://www.parrinst.com/)

Service Engineer

- **Key Projects Handled at Clients' Site:** Reliance, BPCL, IIT's, Delhi university, UOP, Ranbaxy, DRDO (CFEES, DMSRDE), CSIR (IIP, NCL, NPL, IGIB), JUBILANT, and many other research centers and universities.
- Inspection of Instrumentation and Electrical items received at the site.
- Installation and commissioning of reactors and done customization as per client requirements.
- Worked on Spec View automation Software (Tool for SCADA) to control reactor and for data acquisition.
- Worked on transducers and fittings (Swagelok & Parker), mass flow controller and others types of sensors
- Spearheading efforts for scheduling installation at client site to ensure smooth work progress and timely execution of projects with optimum utilization of resources.
- Done troubleshooting analysis which require skills and cost-effective solution.
- Expertise to do **PID to Control the Systems** as per given Set Points.

Achievement

- Design controller for reactor with PC interface to operate and data acquisition.

VIPUL FACILITY MGMT. PVT. LTD

Jan'2010-March'2012

Client: Bharat Petroleum Corporation Ltd, R&D Center, Greater Noida

BPCL R&D pioneered the research on new technologies, catalyst, products development and distillation of crude oil for BPCL that helps India to progress in petroleum field.

<http://www.bharatpetroleum.com/Index.aspx>

Instrumentation Engineer

- Supervision and Execution of routine or non-routine maintenance activities of various Equipments
- Proficient in interpretation of Process Flow Diagrams, P&IDs, Cause and Effect diagrams with Instrument Datasheets of pilot plants.
- Calibration & installation of transmitters such as Pressure transducer, Level Transducer, Flow Transducer, TT & DPT, Thermocouples, RTDs etc.
- Breakdown handling of pilot plants related to fuel extraction using its P&ID and electrical wiring diagram.
- Installed and maintained all the process plant instruments which are generally used to control the plants
- Worked on monitoring equipments to check the frequency or analog output of any transducer like **oscilloscope, signal generator and power supplies**.
- Maintenance of DCS based system with their peripheral devices.

ION EXCHANGE INDIA LTD.

July'09-Nov'10

Instrumentation Engineer

Client: Pragati Power Corporation Ltd., Delhi

- Installation and maintenance of all types of transducers, sensors, valve to control the process plant (**Waste Water Treatment Plant**) etc.
 - Looked out calibration and testing of industrial instruments for precise reading in field.
 - Defined operational parameters and ensuring strict adherence there to generate maximum operational efficiency
 - Done why-why analysis to hit directly on the root cause of breakdown.
 - Resolved the fault with the help of wiring diagram of panels and P&ID of plant.
 - Involved in Process measurement, control and its continuous monitoring.
 - Deftly managed operation and maintenance of RODM plant to convert sewage water into demineralized water under auto mode with the help of PLC/SCADA.
 - Experience on DCS system to monitor and control the plant process activities.
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EDUCATIONAL CREDENTIALS

M. TECH WILP (Automotive Engineering) 2021

Bits Pilani, Hyderabad

Post Graduate Diploma in Industrial Automation, 2013

Prolific Systems and Technology Pvt. Ltd., Noida

B.E (Applied Electronics and Instrumentation), 2009

Arya Institute of Engg and Tech., Jaipur, Rajasthan University

12th, 2004

CBSE board

10th, 2002

CBSE board

Certification

- Certification in Engine testing equipments from M/s Sierra CP, UK.
- Certification in VHDL language to design logic circuits.
- Certification on PLC/SCADA programming from M/s Prolific Systems & Technologies, Noida.

Training Undertaken

- 30 days Training on Introduction, designing, simulation and synthesis electronic circuits through VHDL and implementation of design on FPGA board from Digitech Solutions.
- 45 days Training on IT support (software, hardware, troubleshooting and how to attend IT problems) from BSES Rajdhani Power Ltd, Delhi.

Excellency in Instruments

- **Gas Analyser – CO₂, CO, NO_x, NO, CH₄, THC, O₂AVL, HORIBA**
- **Engine testing equipments –** Dynamometer (eddy current and transient dyno), Fuel and coolant conditioning unit, CAHU, intercooler
- **Automation & Instrumentation –** Transducer or sensors (Pressure, Temperature, Flow, Level, Torque, Load, acceleration), Control valves, Regulators, National Instruments controller, Drives, PLC/SCADA, DCS.
- **Hot and cold chamber equipments –** Compressor, heat exchanger, filter, reservoir, timer
- **PID Controllers –** Delta, Watlow, Honeywell
- **Valves –** Servo valves, motorized valves, pneumatic valves, directional valves.